

Fingerprint Enhancement - Batch Experiment Summary

Method	MSE_mean	PSNR_mean	SSIM_mean	Coherence_mean	Match_mean	Runtime_mean
Degraded input	0.0297	15.8575	0.5016	0.5653	17.5073	0.0
Global HE baseline	0.0355	14.9824	0.5023	0.562	12.2807	0.0035
M1 CLAHE baseline	0.0966	10.3089	0.4489	0.548	3.1207	0.0076
M1 Wiener + CLAHE	0.0481	13.3338	0.567	0.6929	4.4473	0.0121
Gabor	0.078	11.2151	0.4014	0.5728	2.2293	0.9655
Hybrid	0.0461	13.497	0.5569	0.6891	4.1793	0.9481

Dataset mode: SOCOFing Real images with controlled degradation

Example processing pipeline

Clean reference



Low-quality input



Member 1: Wiener + CLAHE



Member 2: Gabor



Member 3: Restored ridges

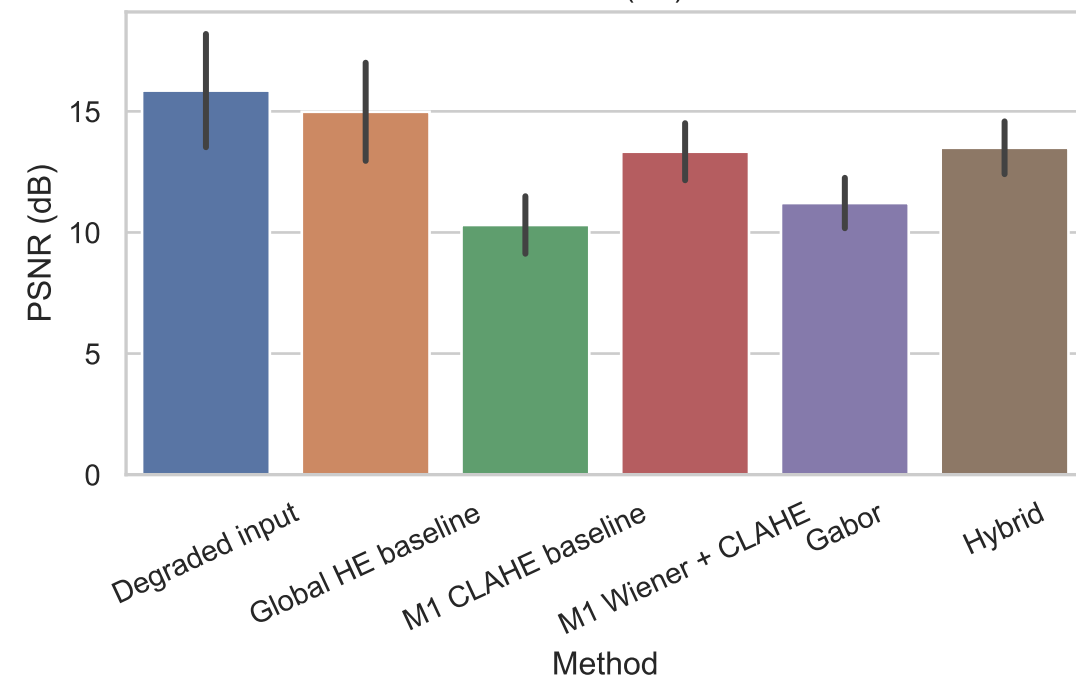


Member 4: Skeleton

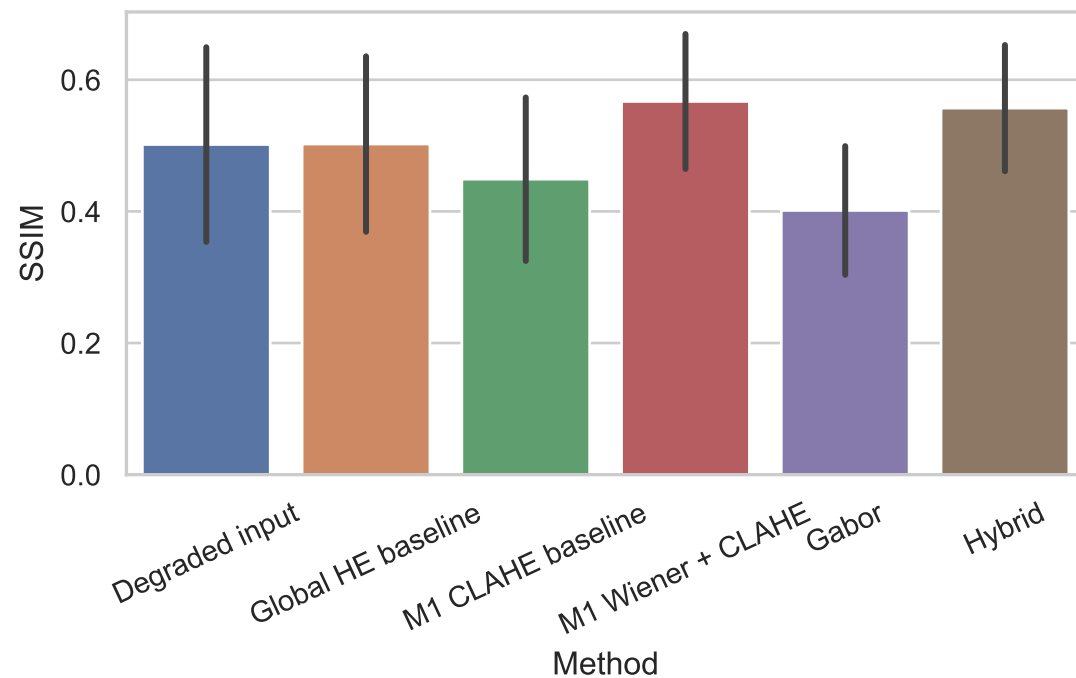


Mean performance with standard deviation

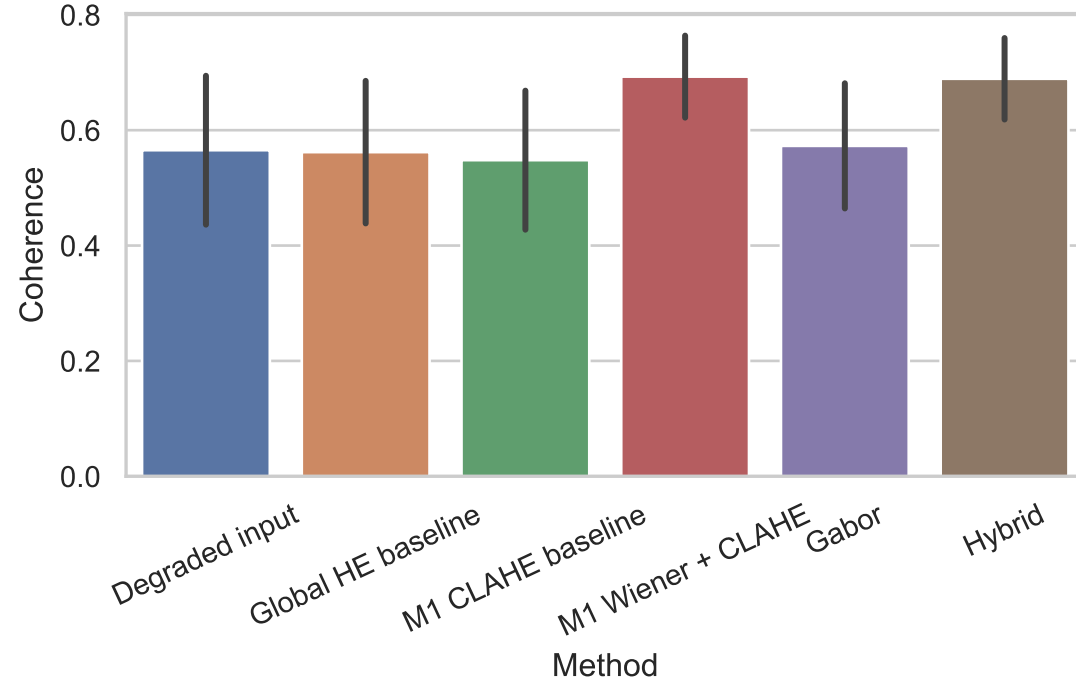
PSNR (dB)



SSIM



Coherence



Match score (%)

